KODBTW / VIVA PREPARATION

Database, persistence and analytics

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This guide follows how user profiles, linked platform accounts and statistics snapshots are modeled and queried. Explain why schema migrations and null-aware analytics protect the meaning of the data.

Contribution reminder: This is a repository-aligned role guide. Describe work you personally performed; confirm individual contribution details with your team.

Explain the system in one flow

Flyway migration	**THEN**	**JPA entities**	**THEN**	**Spring Data repository**	**THEN**	**Service validation / aggregation**	**THEN**	**API response**

Core ideas to explain

Schema ownership

Flyway migrations define database evolution; Hibernate uses `ddl-auto: validate` so it checks mappings without silently changing the schema.

Relationships

Profiles and platform accounts are associated with users. Foreign keys and unique constraints enforce ownership and avoid duplicate account/platform links.

Snapshots

A snapshot stores user, platform, date, source, sync time and available metrics. A unique user/platform/date key supports a daily record.

Null-aware analytics

Unsupported or missing provider fields remain null. Analytics must not treat missing values as real zeroes.

Leaderboard cache

A denormalized cache supports sorted, paginated leaderboard reads; it is refreshed through a restricted admin sync path.

Viva quick preparation

Use these as short answer outlines. Explain the idea in your own words and point to the relevant repository code.

Why use Flyway?

Versioned migrations make schema changes ordered and repeatable across environments. They also provide a clear history of how the schema evolved.

Why snapshot statistics?

Snapshots preserve a time-based record so the app can compare progress and build historical analytics instead of keeping only the latest value.

How is account ownership checked?

Service/repository queries use both the signed-in user ID and account ID before operating on an account. The client does not get to choose another user's scope.

Why keep missing metrics null?

A missing value means "not supplied or not known." Substituting zero would distort totals, comparisons and charts.

Why have a leaderboard cache?

It stores a precomputed read projection to make sorted, paginated responses more efficient. It is derived data and can be rebuilt/refreshed.

Where to point in the repository

```
01 backend/src/main/resources/db/migration/
02 backend/src/main/java/com/kodbtw/entity/Profile.java
03 backend/src/main/java/com/kodbtw/entity/PlatformAccount.java
04 backend/src/main/java/com/kodbtw/entity/PlatformStatSnapshot.java
05 backend/src/main/java/com/kodbtw/repository/
06 backend/src/main/java/com/kodbtw/service/AnalyticsService.java
07 backend/src/main/java/com/kodbtw/entity/LeaderboardCache.java
```

30-second presentation

"The persistence layer uses Flyway for versioned schema changes and JPA for application mappings. Users own profiles and platform accounts, and daily statistics snapshots store the source and timestamp alongside whatever metrics are available. Services scope account access by user ID. Analytics preserve missing metrics as null, while a cache supports efficient leaderboard reads."

Be precise A snapshot is not proof that a provider's value is correct; it records what the provider returned. Distinguish source data, cached data and derived analytics. Never expose another user's private profile data.